

Mathematical Foundations for Computing

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Module 1

Module 1: Sets and Set Theory

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Lesson 1.0 Introduction

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Course structure and navigation

Reading • 15 min

✓

How to learn effectively on this course

Reading • 15 min

✓

Introduction to the specialisation and this course

Video • 4 min

✓

Course Syllabus

Reading • 10 min

Lesson 1.1 Introduction to set theory

✓

The definition of a set

Video • 8 min

✓

The definition of a set

Reading • 10 min

The definition of a set

Practice Assignment • 30 min

The listing method and rule of inclusion

Video • 8 min

The listing method and rule of inclusion

Practice Assignment • 30 min

The powerset of a set

Video • 10 min

The powerset of a set

Practice Assignment • 15 min

Set operations

Video • 10 min

Set operations

Practice Assignment • 30 min

Lesson 1.2 Set representation and manipulation

The representation of a set using Venn diagrams

Video • 4 min

The representation of a set using Venn diagrams

Practice Assignment • 30 min

Mastering Set Operations and Venn Diagram Reasoning

Dialogue • 30 min

De Morgan's laws

Video • 8 min

De Morgan's laws

Practice Assignment • 30 min

Laws of sets: Commutative, associative and distributives

Video • 11 min

Laws of sets: Commutative, associative and distributives

Reading • 15 min

Laws of sets: Commutative, associative and distributive

Practice Assignment • 30 min

Partition of a set

Video • 3 min

Partition of a set

Practice Assignment • 30 min

Lesson 1.3 Summary and Assessment

Module 1 summary

Reading • 15 min

Check your understanding: End of module 1

Graded Assignment • 20 min

Module 2

Module 2: Number Systems and Bases

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Lesson 2.1 Introduction to Number Bases

Introduction to number bases

Video • 13 min

Introduction to number bases – conversion to decimal

Practice Assignment • 30 min

Conversion of integer numbers between different bases

Video • 7 min

Conversion of non-integer numbers between bases

Video • 14 min

Non-integer numbers conversion

Practice Assignment • 30 min

Lesson 2.2 Operations with Binary Numbers

Operations with binary numbers – addition

Video • 3 min

Operations with binary numbers – subtraction

Video • 4 min

Operations with binary numbers – multiplication

Video • 4 min

Number representations

Video • 8 min

Operations with binary numbers

Practice Assignment • 30 min

Why do we need to use these number conversions?

Dialogue • 15 min

Lesson 2.3 Summary and Assessments

Topic 1 – looking back

Video • 2 min

Check your understanding: End of module 2

Graded Assignment • 20 min

Module 3

Module 3: Functions and Their Properties

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Module 4

Module 4: Relations

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Lesson 4.1 Understanding the concept of relations

Definition of a relation: relation versus function

Video • 8 min

Definition of a relation: relation versus function

Practice Assignment • 20 min

Matrix and graph representations of a relation

Video • 10 min

Matrix and graph representations of a relation

Practice Assignment • 20 min

The properties of a relation: reflexive, symmetric and anti-symmetric

Video • 14 min

The properties of a relation: reflexive, symmetric and anti-symmetric

Practice Assignment • 20 min

Relation properties: transitivity

Video • 5 min

The definition of a set

 0:13 / 3:52

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1 Introduction

A set is a fundamental concept in mathematics and computer science. It's a collection of distinct objects, considered as an object in its own right. Sets are used to group related items together and to perform operations on these groups.

2 Definition of a set

Definition 1. A set is a collection of objects. The objects in a set are called its elements or members. The elements in a set can be any types of objects, including sets! The members of a set do not even have to be of the same type. For example, although it may not have any meaningful application, a set can consist of numbers and names.

Definition 2. A set is an unordered collection of distinct elements.
For example: The set of primary colours can be written as {red, blue, yellow}
Elements of a set are also called members.

3 Notation

Sets are usually denoted by capital letters, and their elements are listed within curly braces. For example the set A containing elements 1, 2, and 3 can be written as $A = \{1, 2, 3\}$

3.1 Element or member of set ($\in$)

The symbol $\in$ denotes membership. If a is an element of a set A , we write $a \in A$

For example:

The set $A = \{1, 2, 3\}$ is a set containing the elements 1, 2 and 3.

We write:

$1 \in A$ means 1 is an element of A ,

$2 \in A$ means 2 is an element of A and,

$3 \in A$ means 3 is an element of A .

We refer to not a member of a set using the symbol $\notin$. If a is not an element of a set A , we write $a \notin A$.

For example: let A be a set with $A = \{1, 2, 3\}$. Clearly 4 is not an element of A . We write $4 \notin A$.

3.2 Special sets of numbers

We designate these notations for some special sets of numbers:

$\mathbb{N}$ = the set of natural numbers (positive integers)

$\mathbb{Z}$ = the set of integers

$\mathbb{Q}$ =the set of rational numbers

$\mathbb{R}$ = the set of real numbers

3.3 Empty set ($\emptyset$)

The empty set, also known as the **null set**, is a fundamental concept in set theory and mathematics. It is the set that contains no elements.

Definition 3. The empty set is a set with no elements. It is denoted by $\emptyset$ or $\{\}$. In mathematical notation $\emptyset = \{\}$.

3.4 Cardinality of a set

The cardinality of a set refers to the number of elements in the set. It is a measure of the 'size' of the set.

Definition 4. The cardinality of a set A is denoted by $|A|$, is the number of distinct elements in A .

For example:

For the set $A = \{1, 2, 3, 4\}$, the cardinality $|A| = 4$.

For the set $B = \{a, b, c\}$, the cardinality $|B| = 3$.

For the empty set, $\emptyset$, the cardinality $|\emptyset| = 0$.

3.5 Finite and infinite sets

Finite set: A set with a limited number of elements.

Example: $\{2, 4, 6\}$.

Infinite set: A set with an unlimited number of elements.

Example: The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$.

3.6 Exercise

Using the set notation, list all the elements contained in each of the following sets:

- A = the set of positive integers less than 8.
- B = the set of even positive integers less or equal to 10.
- C = the set of natural numbers.

3.7 Solution

Using the set notation, list all the elements contained in each of the following sets:

- $A = \{1, 2, 3, 4, 5, 6, 7\}$.
- $B = \{2, 4, 6, 8, 10\}$.
- $C = \mathbb{N} = \{1, 2, 3, 4, \dots\}$.

4 Summary

This way of representing sets is called the listing method. In future lessons we will look at more examples of this method as well other set representation methods, such as the the set builder notation method and how sets are represented using Venn diagram.

Go to next item

✓ Completed

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